Shivashriganesh P. Mahato

shivai.dev • smahato@engineering.upenn.edu
ShivashriganeshMahato • shivashriganesh-mahato • shivashriganesh • Google Scholar

BIOGRAPHY

I am a third-year PhD student at the University of Pennsylvania, advised by Prof. Eva L. Dyer, interested in representation learning from neural data. My work spans invasive (intracranial EEG, neural spikes, calcium imaging) and non-invasive (EEG, EMG) neural modalities as I develop new approaches for learning how to decode from brain signals. Of particular interest to me is the challenge of integrating methods across sources of heterogeneity, including temporal drift, subject-to-subject variability, and diverse behaviors. I am interested in developing flexible tools that can enable clinical outcomes downstream, empower assistive technologies, and hopefully enhance our understanding of the brain.

EDUCATION

PhD in Computer and Information Science Aug 2024 – Present
University of Pennsylvania (transferred from Georgia Tech) • Advisor: Prof. Eva L. Dyer

- Relevant coursework: ML for Neural/Behavioral Data, Systems Neuroscience, Discrete Generative Models

MS in Computer Science Aug 2022 – Aug 2023
Texas A&M University • Advisor: Prof. Yoonsuck Choe

- Relevant coursework: Theory of Deep Learning, ML, Operating Systems, Info Retrieval, Stochastic Processes
- Thesis: *Comparison of Evolved Memory Mechanisms in Neural Networks*

BS in Applied Mathematics Aug 2019 – May 2022
Texas A&M University

- Minors in Computer Science and Statistics, graduated Summa Cum Laude with Honors
- Relevant coursework: Topological Data Analysis, Multivariate Analysis, Statistical Learning, Applied Analytics

PUBLICATIONS

** denotes co-first authors*

- **Mahato, S.P.***, Balaji, K.*, Lachi, D., Dyer, E.L. "Population-Aware Contrastive Channel Embeddings for Intracranial EEG." Under review at *2027 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*.
- Andre, A.*, **Mahato, S.P.***, Arora, V., Balaji, K., Lachi, D., Krishna, N.H., Xiao, J., Zhang, Y., Mao, X., Ma, W., Yu, H., International Brain Laboratory, Birman, D., Bonacchi, N., Chapuis, G.A., Catarino, J.A., Davatolhagh, F., Faulkner, M., Freitas-Silva, L., Hu, F., Huntenburg, J.M., Khanal, A., Laranjeira, I., Lau, P., Meijer, G.T., Miska, N.J., Noel, J.-P., Pan-Vazquez, A., Raiser, G., Rossant, C., Socha, K.Z., Urai, A.E., Wells, M.J., West, S.J., Winter, O., Richards, B., Lajoie, G., Hurwitz, C., Azabou, M., Whiteway, M.R., Paninski, L., Dyer, E.L. (2026). BrainWideBench: Benchmarking Large-Scale Pretraining and Across-Animal Transfer in Multi-Region Neural Recordings. *arXiv preprint arXiv:2609.22064*. Under review at *ICLR 2027*.
- Lachi, D., Arora, V., **Mahato, S.P.**, Andre, A., Dyer, E.L. (2026). "Scaling Neural Motor Decoding via Decoupled Behavioral Pretraining." To appear in *2026 Neural Information Processing Systems (NeurIPS)*.
- Arora, V., Lachi, D., **Mahato, S.P.**, Andre, A., Poschl, J., Chapuis, G., Whiteway, M.R., Dyer, E.L. (2026). "Zero-Shot Across-Subject Neural Decoding via Neuron-Level Representations." To appear in *NeurIPS 2026 Workshop on Symmetry and Geometry in Neural Representations (NeurReps)*, Extended Abstract. Full version under review at *ICLR 2027*.
- **Mahato, S.P.***, Xiao, J.*, Andre, A., Chau, G., Ma, W., Knight, I.J., Nguyen, D., Hu, L.J., Brunton, B.W., Beauchamp, M.S., Pesaran, B., Shuvaev, S.A., Dyer, E.L. (2025). A scalable self-supervised method for modeling human intracranial recordings during natural behavior. *NeurIPS 2025 Workshop on Foundation Models for the Brain and Body.* Spotlight (top 15%).
- Arora, V.*, Lachi, D.*, **Mahato, S.P.**, Azabou, M., Chen, Z., Dyer, E.L. (2025). Exploiting All Laplacian Eigenvectors for Node Classification with Graph Transformers. *NeurIPS 2025 Workshop on New Perspectives in Graph Machine Learning*.
- Lachi, D., Mohammadi, M., Meyer, J., Arora, V., **Mahato, S.P.**, Palczewski, T., Dyer, E.L. (2025). Learning from Multi-Table Relational Data with the Relational Graph Perceiver. *EurIPS 2025 Workshop on AI for Tabular Data*.
- Azabou, M., Arora, V., Sobral, M., Suarez, L., Krishna, N., Ryoo, A., Mao, X., **Mahato, S.P.**, Andre, A., Paninski, L., Lajoie, G., Richards, B., Dyer, E.L. (2025). "An Open-Source Ecosystem for Models of Multi-Modal Brain and Body

Data." *NeurIPS 2025 Workshop on Foundation Models for the Brain and Body*, Demo Track.

- **Mahato, S.P.**, Kaliyur, S., Chung, J.R., Choe, Y. (2024). Use of External Markers by Reactive Agents as an Easier Evolutionary Route Toward Memory. *2024 International Joint Conference on Neural Networks (IJCNN)* (pp. 1–9). IEEE.

RESEARCH EXPERIENCE

Graduate Research Assistant (PhD), Neural Data Science (NerDS) Lab Aug 2024 – Present
PI: Prof. Eva L. Dyer • University of Pennsylvania

- Created and deployed a new benchmark for neuro-foundation models on the IBL Brainwide Map. We setup an evaluation suite across three task domains (decoding, neural activity prediction, and anatomical localization) for large-scale learning from neural spiking activity. We rigorously compared classical models and state-of-the-art foundation models to serve as standard baselines for the community (in review, NeurIPS 2026).
- Developed a self-supervised learning framework for large-scale intracranial EEG, which reconstructs masked channels to learn general neural representations that improve downstream decoding performance (NeurIPS Workshop 2025).
- Co-authored works on graph learning, including a principled method for frequency-informed selection of Laplacian eigenvector positional embeddings in Graph Transformers, and large-scale graph learning on dynamic relational data (NeurIPS Workshop 2025).

Graduate Researcher (MS Thesis), Brain Networks Lab Jan 2023 – Aug 2023
PI: Prof. Yoonsuck Choe • Texas A&M University

- Studied the evolution of memory in neural controllers in an empirical testbed, focusing on how memory could evolve from fully reactive agents and highlighting the feasibility of an external-to-internal route towards memory development to qualify previous works (IJCNN 2024).
  - Under a simulated neuroevolution framework (NEAT), we trained different memory-capable systems (RNN, TDNN, FFN with external interactive capabilities [FFWD]) in a memory-dependent RL task.
  - We then analyzed network topologies, developed a novel characterization of evolutionary efficiency, and qualitatively compared dynamics in-distribution as well as with perturbed environmental conditioning to demonstrate that FFWD is a rudimentary memory mechanism.

REU Scholar, Department of Statistics Apr 2022 – Aug 2022
PI: Prof. Sharmista Guha • Texas A&M University

- Compared multiple neuroimaging datasets for lab initiative towards modeling clinical disease progression, selecting OASIS-3 for its multimodal imaging and cognitive (Alzheimer's) assessments.
- Developed a preprocessing pipeline (registration of Diffusion Tensor Imaging to T1 space, tractography analysis, generation of adjacency matrices) to obtain input features for Bayesian Network Regression.

REU Scholar, Department of Mathematics May 2020 – Jul 2020
PI: Prof. Wencai Liu • Texas A&M University

- Contributed to a re-formulation of the standard solution to the Ambarzumian problem with Dirichlet boundary conditions, framing it as an inverse spectral problem.

Student Research Assistant, Texas A&M AgriLife Oct 2019 – Mar 2020
PI: Prof. Muthukumar Bagavathiannan • Texas A&M University

- Performed extensive data annotation on crop field imagery captured by UAV, yielding a large-scale supervised dataset with segmentation labels.
- Tested various feature extraction schemes and trained a preliminary model (finetuned U-Net-ResNet50) for semantic segmentation of weeds and crops, achieving 90% pixel-wise classification accuracy.

PROFESSIONAL EXPERIENCE

Capital One Chicago, IL

Senior Associate Software Engineer Jul 2024 – Aug 2024

- Enhanced observability of universal payments infrastructure through end-to-end traceability instrumentation and improved metrics collection and logging.

Associate Software Engineer Aug 2023 – Jul 2024

- Led design, implementation, and full-stack testing (E2E, performance, resiliency) of a major payments scheduling feature; established team-wide performance testing standards as SME.
- Architected and deployed an automated email reporting service for payments fulfillment reconciliation, customized per line of business.
- Authored AWS cloud compliance procedures and on-call troubleshooting runbooks, codifying operational knowledge for team-wide use.

Sellers Shield Austin, TX

Software Engineer II Jan 2023 – Mar 2023

- Standardized backend testing infrastructure and led OAuth implementation for the partner-facing API; implemented UI wireframes for platform migration.

Capital One Plano, TX

Software Engineering Intern Jun 2022 – Aug 2022

- Built and deployed a serverless AWS web application for audit log validation during platform migration, cutting the validation process from days to hours while eliminating downtime costs.

Ascension Digital Studio Austin, TX

Software Engineering Intern May 2021 – Aug 2021

- Reduced test time by ∼45% by embedding a Mongo database into the test suite; developed and shipped features for a surgical scheduling system with production deployments on GCP.
- Provisioned the surgical scheduling manager on the enterprise service layer to enable smooth migration from legacy EHR systems.

Vitalus Health Katy, TX

Contract Software Developer Feb 2021 – May 2021

- Implemented a general-purpose SMS API serving as the communication backbone for all customer-facing interactions.
- Built custom logging, a patient scheduling system, and a data provisioning layer supporting rapid migration and NLP chatbot integration.

TEACHING EXPERIENCE

Teaching Assistant, CIS 7000: Deep Learning for Time Series (DL4TS) Spring 2026
University of Pennsylvania

Teaching Assistant, ECE 4252/8803: Fundamentals of Machine Learning Fall 2024
Georgia Institute of Technology

CONFERENCE & WORKSHOP PARTICIPATION

- Presented "Measuring Progress in Neuro Foundation Models with BrainWideBench" at an *International Brain Labratory (IBL) lab meeting* (October 2026).
- Attended the *2026 ARNI Annual Retreat* and presented "Measuring Progress in Neuro Foundation Models with BrainWideBench". New York City, NY, USA (September 2026).
- Presented "BrainWideBench: Benchmarking large-scale pretraining on multi-region neural recordings" at the *ARNI Working Group on Neuro-Foundation Models*. New York City, NY, USA (February 2026).
- Presented "A scalable self-supervised method for modeling human intracranial recordings during natural behavior" at the *NeurIPS 2025 Workshop on Foundation Models for the Brain and Body* (oral and poster). San Diego, CA, USA (December 2025).
- Attended the *2025 CCN Workshop: "Functional alignment of information encoded in neural activity"* (Dartmouth). Hanover, NH, USA (September 2025).
- Presented "Large-Scale Deep Learning in Neuroscience" at the *Computational Neural-Engineering Training Program Tutorial Series* (Georgia Tech). Atlanta, GA, USA (June 2025).
- Presented "Machine Learning for Adaptive DBS" at the *Adaptive Deep Brain Stimulation Workshop* (University of Pennsylvania). Philadelphia, PA, USA (April 2025).
- Presented "Large-Scale Deep Learning in Neuroscience" at the *Data-Driven and Large-Scale Modeling in Neuroscience Workshop* (Georgia Tech). Atlanta, GA, USA (April 2025).

- TAed "Transformers in Neuroscience" Tutorial at the *Computational and Systems Neuroscience (COSYNE)*. Montreal, Canada (March 2025).
- Presented "Use of External Markers by Reactive Agents as an Easier Evolutionary Route Toward Memory" at the *IEEE World Conference on Computational Intelligence (WCCI)*. Yokohama, Japan (June 2024).

HONORS, AWARDS & CERTIFICATIONS

- Top Reviewer, NeurIPS (2026)
- President's Fellowship, Georgia Institute of Technology (2024)
- IEEE Computational Intelligence Society (CIS) "Industry Early Career" Participation and Travel Grant (2024)
- AWS Certified Solutions Architect, Associate (May 2024 – May 2027)
- Honors Degree in Mathematics, Texas A&M Department of Mathematics (2022)
- Certificate of Excellence, Texas A&M Department of Mathematics (2022)
- Classroom Excellence Award for MATH 470 (Cryptography), Texas A&M Department of Mathematics (2021)

PROFESSIONAL AFFILIATIONS

- Reviewer, Neural Information Processing Systems (*NeurIPS*) (2026)
- Reviewer, Computational and Systems Neuroscience (*COSYNE*) (2026, 2027)
- Reviewer, NeurIPS 2025 Workshop on Foundation Models for the Brain and Body (2025, 2026)
- Reviewer, International Joint Conference on Neural Networks (*IJCNN*) (2024, 2025)
- Reviewer, IEEE Transactions on Cognitive and Developmental Systems (2024, 2025)
- Member, Institute of Electrical and Electronics Engineers (2024 – 2025)
- Member, IEEE Computational Intelligence Society (2024 – 2025)

MENTORSHIP

- Keshav Balaji, MS Computer and Information Science, University of Pennsylvania (Ongoing)
- Reid Laughton, MS Computer and Information Science, University of Pennsylvania (2026)
- Jake Manning, MS Computer Science, Texas A&M University (2025)
- Tanu Shree, MS Computer Science, Texas A&M University (2023)
  - Thesis: *Classification of Cricket Songs Based on Their Genus and Species*

TECHNICAL SKILLS

- **ML Tools:** pytorch, lightning, numpy, cupy, pandas, sklearn, keras, neat-python, wandb, slurm
- **Languages:** Python, Java, C++, Javascript, R, SQL
- **Web Tools:** Spring Boot, Django, NodeJS, Flask, React, VueJS
- **Dev Tools:** Git, Docker, Kubernetes, Linux, Jenkins, Splunk, New Relic, Jira, Terraform
- **Cloud:** AWS, Google Cloud Platform (GCP), Firebase, Cloudflare